Auteur: Thijs Van Schel
Studierichting: Informatica
Oefeningenreeks 4A nr. 47.5

$$\int \frac{2x^5+6x^3-2x}{x^6-3x^2-2}dx = \int \frac{2x(x^4+3x^2-1)}{x^6-3x^2-2}dx$$

$t = x^2 \Rightarrow dt = 2xdx$

$$\int \frac{t^2+3t-1}{t^3-3t-2}dt = \int \frac{t^2+3t-1}{(t+1)(t^2-t-2)}dt$$

$$= \int \frac{t^2+3t-1}{(t+1)^2(t-2)}dt \Rightarrow \frac{A}{(t+1)^2} + \frac{B}{(t+1)} + \frac{C}{(t-2)}$$

$$A = \left.\frac{t^2+3t-1}{t-2}\right|_{x=-1} = 1$$

$$C = \left.\frac{t^2+3t-1}{(t+1)^2}\right|_{x=2} = 1$$

$$\frac{t^2+3t-1}{(t+1)^2(t-2)} - \frac{1}{(t+1)^2} = \frac{t^2+3t-1-(t-2)}{(t+1)^2(t-2)}$$

$$= \frac{t^2+2t+1}{(t+1)^2(t-2)} = \frac{(t+1)^2}{(t+1)^2(t-2)} = \frac{1}{(t-2)}$$

$$B = \frac{\frac{1}{t-2}}{t+1} = \left.\frac{t+1}{t-2}\right|_{x=-1} = 0$$

$$\int \frac{1}{(t+1)^2}dt + \int \frac{1}{t-2}dt = -\frac{1}{t+1} + \ln|t-2| + C$$

$$= -\frac{1}{x^2+1} + \ln|x^2-2| + C$$

References